

Daniel Thorngren

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Johns Hopkins University
Physics and Astronomy Department
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Education **University of California, Santa Cruz (2013-2019)**
Ph.D. in Physics (Advisor: Jonathan Fortney)
Master of Science in Physics (2015)

University of California, Davis (2008-2013)
Bachelor of Science in Physics, Highest Honors (Advisor: Mani Tripathi)

Research Interests Giant planets - composition, structure evolution, thermal transport, anomalous heating, core physics, interior-atmosphere interactions.
Astrostatistics - Bayesian modeling of astrophysical data and populations.
Planet formation and its effect on observable outcomes

Skills Constructing mathematical models of physical systems
Statistical modeling and inference - generalized linear models, parametric inference, Gaussian processes, and hierarchical Bayesian models.
Machine learning techniques - PCA, SVM, and neural networks.
Programming in C, C++, C#, Python, Cython, Numba, and R.
Data analysis tools - SQL, Scipy, Matplotlib, Pandas.
High-performance computing, working in a Unix environment
Implementing advanced MCMC techniques (e.g. Hamiltonian Monte Carlo, BPS).

Experience **Davis Fellowship** - Postdoctoral Researcher, Johns Hopkins University (2022-present)
Modeling the mass loss and tidal circularization of exoplanets.
Supporting observational studies of exoplanets with interior structure models.
Developing new retrievals for the interior structures of directly-imaged planets.

Trottier Fellowship - Postdoctoral Researcher, University of Montréal (2019-2022)
Modeled the effect of evolving main-sequence stars on their planets' radii.
Modeled the mass loss of hot Saturn-mass exoplanets.
Studied the effects of tidal circularization on giant planet interior structure.
Outreach talk to transitional Kindergartners at Oliveira Elementary School

Other Worlds Laboratory (OWL) - Graduate Student, UCSC (2013-2019)
Created computer models of the interior structure of giant planets
Analyzed observed exoplanet data to infer the bulk composition of giant exoplanets and understand their relationship with mass through Bayesian statistics
Analyzed stellar abundance data for the connection with planetary composition
Assisted with MCMC-based inference of planetary interior structures using gravity moment measurements from the Cassini and Juno spacecraft.
Advised two undergraduates for their senior theses.
Taught lab sections for introductory physics classes and graded coursework.

Amazon A9 - Applied Science Intern (June 2018 - September 2018)
Constructed machine-learned models for ranking customer search results, which were subsequently deployed to the Amazon website.
Analyzed Amazon internal books data to construct new features for use in improving the quality of customer search results.

LUX Dark Matter Detector - Undergraduate Researcher, UC Davis (2011-2013)
Analyzed LUX data and calculated expected detection rates
Wrote, modified, and operated particle physics simulation software
Created optimized approximate simulation module for many-photon events
which improved run-time by a factor of > 100 (senior thesis work)

Honors Member of Sigma Pi Sigma (Society of Physics Students honors society)
Dean's list five times
Highest Honors from UC Davis for senior thesis work
Trottier Postdoctoral Fellowship 2019
Davis Postdoctoral Fellowship 2022

Software **Planet Slicer** – Package for fitting brightness maps to phase curves of reflective
or self-luminous objects. github.com/dpthorngren/PlanetSlicer
Sam – Flexible MCMC sampling package written in Cython for
difficult astrophysical use cases. github.com/dpthorngren/Sam
Eggman – Python package for computing the transit curves of piecewise-ellipsoidal
objects (limb asymmetry, tidal distortion). github.com/dpthorngren/Eggman
Starlord – Python package for fitting to stellar observations, flexibly varying grid
and parameter choice (e.g. $[\alpha/\text{Fe}]$). github.com/dpthorngren/Starlord

Talks and Presentations

1. Exoclimates VII - Atmospheres to Cores: Giant Planets with JWST (7/9/25)
2. Carnegie Earth and Planetary Laboratory NCAD Seminar
Observational Constraints on Runaway Gas Accretion (3/21/25)
3. Max Planck Institute for Astronomy Exoplanet Seminar
Exoplanet Interiors in the JWST Era (6/24/24)
4. Exoplanets V Plenary Talk
Exoplanet Interiors in the JWST Era (6/18/24)
5. Imperial College London Exoplanets Seminar
Exoplanet Interiors in the JWST Era & Hot Saturn Runaway Mass Loss (6/13/24)
6. Oxford SPIMAX Seminar
Exoplanet Interiors in the JWST Era (6/11/24)
7. NASA Goddard SEEC Non-Transiting Planets Symposium
The Deep Interiors of Directly Imaged Planets (4/16/24)
8. Carnegie Earth & Planetary Laboratory Astronomy Seminar
Exoplanet Deep Interiors With JWST (3/15/24)
9. UC Santa Cruz Flash Seminar
Exoplanet Deep Interiors With JWST (3/1/24)
10. UC Berkeley TAC Seminar
Exoplanet Deep Interiors With JWST (2/26/24)
11. Pennsylvania State University CEHW Seminar
Exoplanet Deep Interiors With JWST (2/5/24)
12. Open Problems in the Astrophysics of Gas Giants (OPAGA) Conference (12/5/23)
Synergy Between Atmospheric and Bulk Studies of Giant Planets
13. STScI 2023 Spring Symposium (5/16/23)
Using JWST Spectra to Explore Exoplanet Deep Interiors
14. Princeton Astrophysics Department Seminar (10/10/22)
Formation and Mass Loss of Exo-Saturns
15. University of Maryland Exoplanet Seminar (9/28/22)
Hot Saturns: Formation and Mass Loss
16. Math+X Symposium (7/11/22)
Giant Planet Population Physics

17. TESS Science Conference 2 (8/3/21)
Exoplanet Interior Physics in the TESS Era (Invited Talk & Panel)
18. JWST Early Release Science Program Workshop (7/1/21)
What Masses and Radii Tell us About Planets (Review Talk)
19. Canada Planet Discussion Day (6/10/21)
Giant Exoplanet Interiors (Review Talk)
20. American Astronomical Society Meeting (6/9/21)
The Diverse Hot Saturn Population: Composition, Thermal Evolution, and Mass Loss
21. NASA Goddard SFC Exoplanet Seminar (1/6/21)
Slow Cooling and Fast Reinflation for Hot Jupiters
22. Chesapeake Bay Area Exoplanet Meeting (12/11/20)
Slow Cooling and Fast Reinflation for Hot Jupiters
23. PLATO Extra-Solar Planet Workshop (11/30/20)
Slow Cooling and Fast Reinflation for Hot Jupiters
24. NExSci Exoplanet Demographics Conference (11/10/20)
Giant Planet Population Physics (Invited Review Talk)
25. Caltech Division of Geological and Planetary Sciences Seminar (6/4/19)
Giant Exoplanet Physics From Population Statistics
26. American Astronomical Society Meeting (1/10/19)
Bayesian Inference of Giant Exoplanet Physics (Thesis Talk)
27. AAS Division of Planetary Science (10/24/18)
Bayesian Inference of Giant Planet Physics (Thesis Talk)
28. Bay Area Exoplanets Meeting (6/1/18)
Giant Exoplanet Main Sequence Reinflation & Atmosphere Metallicity
29. MIT Kavli Institute Exoplanet Tea Talk (4/4/18)
Bayesian Inference of Giant Planet Physics
30. Harvard-Smithsonian Center for Astrophysics Stars and Planets Seminar (4/2/18)
Bayesian Inference of Giant Planet Physics
31. American Astronomical Society Meeting (1/10/18)
Bayesian Inference of Hot Jupiter Radii: Evidence for Ohmic Dissipation?
32. AAS Division of Planetary Sciences Meeting (10/19/17)
Bayesian Inference of Hot Jupiter Radii: Evidence for Ohmic Dissipation?
33. Exoclipse Conference, Boise (8/21/17)
Bayesian Inference of Hot Jupiter Radii Points to Ohmic Dissipation
34. American Astronomical Society Meeting (1/5/17)
Bayesian Inference of Giant Planet Physics
35. Bay Area Exoplanets Meeting (12/9/16)
Bayesian Inference of Giant Planet Physics
36. AAS Division of Planetary Sciences Meeting (10/17/16)
Bayesian Inference of the Composition and Inflation Power of Hot Jupiters
37. Giant Magellan Telescope Meeting (9/26/16)
Bayesian Inference of Giant Planet Physics (Poster)
38. Linking Exoplanet and Disk Compositions, Space Telescope Science Institute (9/12/16)
Examining the Bulk Metallicity of Giant Planets
39. Exoplanets I Meeting (7/3/16)
Giant Planet Composition and Inflation: Breaking the Degeneracy (Poster)
40. Extreme Solar Systems Meeting (11/29/15) - The Metallicity of Giant Planets (Poster)
41. Bay Area Exoplanets Meeting (9/30/15) - The Metallicity of Giant Planets

Publications – 8 First Author, 66 total, H-index: 29

First Author and Select Publications

1. Thorngren, D. P., Fortney, J. J., Murray-Clay, R. A., Lopez, E. D. (2016)
The Mass-Metallicity Relation for Giant Planets
The Astrophysical Journal; 1, 64. 385 citations.
2. Thorngren, D. P., Fortney, J. J. (2018)
Bayesian Analysis of Hot-Jupiter Radius Anomalies: Evidence for Ohmic Dissipation?
The Astronomical Journal; 5, 214. 178 citations.
3. Thorngren, D., Gao, P., Fortney, J. J. (2019)
The Intrinsic Temperature and Radiative-Convective Boundary Depth in the Atmospheres of Hot Jupiters
The Astrophysical Journal; 1, L6. 115 citations.
4. Thorngren, D., Fortney, J. J. (2019)
Connecting Giant Planet Atmosphere and Interior Modeling: Constraints on Atmospheric Metal Enrichment
The Astrophysical Journal; 2, L31. 110 citations.
5. Thorngren, D. P., Fortney, J. J., Lopez, E. D., Berger, T. A., et al. (2021)
Slow Cooling and Fast Reinflation for Hot Jupiters
The Astrophysical Journal; 1, L16. 43 citations.
6. Thorngren, D. P., Lee, E. J., Lopez, E. D. (2023)
Removal of Hot Saturns in Mass-Radius Plane by Runaway Mass Loss
The Astrophysical Journal; 2, L36. 27 citations.
7. Thorngren, D. P. (2024)
The Hot Jupiter Radius Anomaly and Stellar Connections
Handbook of Exoplanets, 2nd Edition (in editing); arXiv:2405.05307. 10 citations.
8. Thorngren, D. P., Sing, D. K., & Mukherjee, S. (2025)
Bayesian Model Comparison and Significance: Widespread Errors and how to Correct Them
Accepted to ApJS, arXiv e-prints; arXiv:2510.00169
9. Gao, P., Thorngren, D. P., Lee, E. K. H., Fortney, J. J., et al. (2020)
Aerosol composition of hot giant exoplanets dominated by silicates and hydrocarbon hazes
Nature Astronomy; 951. 210 citations.
10. Fortney, J. J., Visscher, C., Marley, M. S., Hood, C. E., et al. (2020)
Beyond Equilibrium Temperature: How the Atmosphere/Interior Connection Affects the Onset of Methane, Ammonia, and Clouds in Warm Transiting Giant Planets
The Astronomical Journal; 6, 288. 112 citations.
11. Kreidberg, L., Line, M. R., Thorngren, D., Morley, C. V., et al. (2018)
Water, High-altitude Condensates, and Possible Methane Depletion in the Atmosphere of the Warm Super-Neptune WASP-107b
The Astrophysical Journal; 1, L6. 91 citations.
12. Yadav, R. K., Thorngren, D. P. (2017)
Estimating the Magnetic Field Strength in Hot Jupiters
The Astrophysical Journal; 1, L12. 82 citations.
13. Lee, E. J., Karalis, A., Thorngren, D. P. (2022)
Creating the Radius Gap without Mass Loss
The Astrophysical Journal; 2, 186. 64 citations.
14. Sing, D. K., Rustamkulov, Z., Thorngren, D. P., Barstow, J. K., et al. (2024)
A warm Neptune's methane reveals core mass and vigorous atmospheric mixing
Nature; 8018, 831. 59 citations.

Remaining Publications by Date

15. Brahm, R., Trifonov, T., Jordán, A., Henning, T., et al. (2025)
A Compact Multi-Planet System of Three Transiting Giant Planets Around TIC118798035
arXiv e-prints; arXiv:2510.19162. Submitted to AAS Journals; in revision.
16. Schmidt, S. P., MacDonald, R. J., Tsai, S.-M., Radica, M., et al. (2025)
A Comprehensive Reanalysis of K2-18 b's JWST NIRISS+NIRSpec Transmission Spectrum
The Astronomical Journal; 6, 298.
17. Tang, Y., Fortney, J. J., Nimmo, F., Thorngren, D., et al. (2025)
Reassessing Sub-Neptune Structure, Radii, and Thermal Evolution
The Astrophysical Journal; 1, 28. 7 citations.

18. Bryant, E. M., Jordán, A., Hartman, J. D., Bayliss, D., et al. (2025)
A transiting giant planet in orbit around a 0.2-solar-mass host star
Nature Astronomy; 1031. 3 citations.
19. Fu, G., Stevenson, K. B., Sing, D. K., Mukherjee, S., et al. (2025)
Statistical Trends in JWST Transiting Exoplanet Atmospheres
The Astrophysical Journal; 1, 1. 10 citations.
20. Wang, G., Balmer, W. O., Pueyo, L., Thorngren, D., et al. (2025)
A Revised Density Estimate for the Largest Known Exoplanet, HAT-P-67 b
The Astronomical Journal; 6, 336.
21. Chachan, Y., Dalba, P. A., Thorngren, D. P., Kane, S. R., et al. (2025)
Giant Outer Transiting Exoplanet Mass (GOT 'EM) Survey. V. Two Giant Planets in Kepler-511 but Only One Ran Away
The Astronomical Journal; 5, 248. 1 citations.
22. Yee, S. W., Stefánsson, G., Thorngren, D., Monson, A., et al. (2025)
The Super-puff WASP-193 b is on a Well-aligned Orbit
The Astronomical Journal; 4, 225. 1 citations.
23. Kirk, J., Ahrer, E.-M., Claringbold, A. B., Zamyatina, M., et al. (2025)
BOWIE-ALIGN: JWST reveals hints of planetesimal accretion and complex sulphur chemistry in the atmosphere of the misaligned hot Jupiter WASP-15b
Monthly Notices of the Royal Astronomical Society; 4, 3027. 20 citations.
24. Karalis, A., Lee, E. J., Thorngren, D. P. (2025)
Separating Super-puffs versus Hot Jupiters among Young Puffy Planets
The Astrophysical Journal; 1, 46. 6 citations.
25. Balmer, W. O., Franson, K., Chomez, A., Pueyo, L., et al. (2025)
VLTI/GRAVITY Observations of AF Lep b: Preference for Circular Orbits, Cloudy Atmospheres, and a Moderately Enhanced Metallicity
The Astronomical Journal; 1, 30. 17 citations.
26. Thao, P. C., Mann, A. W., Feinstein, A. D., Gao, P., et al. (2024)
The Featherweight Giant: Unraveling the Atmosphere of a 17 Myr Planet with JWST
The Astronomical Journal; 6, 297. 25 citations.
27. Morley, C. V., Mukherjee, S., Marley, M. S., Fortney, J. J., et al. (2024)
The Sonora Substellar Atmosphere Models. III. Diamondback: Atmospheric Properties, Spectra, and Evolution for Warm Cloudy Substellar Objects
The Astrophysical Journal; 1, 59. 86 citations.
28. Swain, M. R., Hasegawa, Y., Thorngren, D. P., Roudier, G. M. (2024)
Planet Mass and Metallicity: The Exoplanets and Solar System Connection
Space Science Reviews; 6, 61. 13 citations.
29. Nabbie, E., Huang, C. X., Burt, J. A., Armstrong, D. J., et al. (2024)
Surviving in the Hot-Neptune Desert: The Discovery of the Ultrahot Neptune TOI-3261b
The Astronomical Journal; 3, 132. 10 citations.
30. Grunblatt, S. K., Saunders, N., Huber, D., Thorngren, D., et al. (2024)
TESS Giants Transiting Giants. IV. A Low-density Hot Neptune Orbiting a Red Giant Star
The Astronomical Journal; 1, 1. 9 citations.
31. Vissapragada, S., Greklek-McKeon, M., Linssen, D., MacLeod, M., et al. (2024)
Helium in the Extended Atmosphere of the Warm Superpuff TOI-1420b
The Astronomical Journal; 5, 199. 9 citations.
32. Dalba, P. A., Kane, S. R., Isaacson, H., Fulton, B., et al. (2024)
Giant Outer Transiting Exoplanet Mass (GOT 'EM) Survey. IV. Long-term Doppler Spectroscopy for 11 Stars Thought to Host Cool Giant Exoplanets
The Astrophysical Journal Supplement Series; 1, 16. 5 citations.
33. Radica, M., Coulombe, L.-P., Taylor, J., Albert, L., et al. (2024)
Muted Features in the JWST NIRISS Transmission Spectrum of Hot Neptune LTT 9779b
The Astrophysical Journal; 1, L20. 45 citations.
34. Pereira, F., Grunblatt, S. K., Psaridi, A., Campante, T. L., et al. (2024)
TESS giants transiting giants V - two hot Jupiters orbiting red giant hosts
Monthly Notices of the Royal Astronomical Society; 3, 6332. 8 citations.
35. Eberhardt, J., Hobson, M. J., Henning, T., Trifonov, T., et al. (2023)
Three Warm Jupiters around Solar-analog Stars Detected with TESS
The Astronomical Journal; 6, 271. 12 citations.

36. Mann, C. R., Dalba, P. A., Lafrenière, D., Fulton, B. J., et al. (2023)
Giant Outer Transiting Exoplanet Mass (GOT 'EM) Survey. III. Recovery and Confirmation of a Temperate, Mildly Eccentric, Single-transit Jupiter Orbiting TOI-2010
The Astronomical Journal; 6, 239. 8 citations.
37. Yoshida, S., Vissapragada, S., Latham, D. W., Bieryla, A., et al. (2023)
TESS Spots a Super-puff: The Remarkably Low Density of TOI-1420b
The Astronomical Journal; 5, 181. 11 citations.
38. Bean, J. L., Xue, Q., August, P. C., Lunine, J., et al. (2023)
High atmospheric metal enrichment for a Saturn-mass planet
Nature; 7963, 43. 47 citations.
39. Narang, M., Oza, A. V., Hakim, K., Manoj, P., et al. (2023)
uGMRT observations of the hot-Saturn WASP-69b: Radio-Loud Exoplanet-Exomoon Survey II (RLEES II)
Monthly Notices of the Royal Astronomical Society; 2, 1662. 14 citations.
40. Calissendorff, P., De Furio, M., Meyer, M., Albert, L., et al. (2023)
JWST/NIRCam Discovery of the First Y+Y Brown Dwarf Binary: WISE J033605.05-014350.4
The Astrophysical Journal; 2, L30. 24 citations.
41. Piaulet, C., Benneke, B., Almenara, J. M., Dragomir, D., et al. (2023)
Evidence for the volatile-rich composition of a 1.5-Earth-radius planet
Nature Astronomy; 206. 67 citations.
42. Greklek-McKeon, M., Knutson, H. A., Vissapragada, S., Jontof-Hutter, D., et al. (2023)
Constraining the Densities of the Three Kepler-289 Planets with Transit Timing Variations
The Astronomical Journal; 2, 48. 12 citations.
43. Narang, M., Oza, A. V., Hakim, K., Manoj, P., et al. (2023)
Radio-loud Exoplanet-exomoon Survey: GMRT Search for Electron Cyclotron Maser Emission
The Astronomical Journal; 1, 1. 20 citations.
44. Komacek, T. D., Gao, P., Thorngren, D. P., May, E. M., et al. (2022)
The Effect of Interior Heat Flux on the Atmospheric Circulation of Hot and Ultra-hot Jupiters
The Astrophysical Journal; 2, L40. 18 citations.
45. Jacobs, B., Désert, J.-M., Pino, L., Line, M. R., et al. (2022)
A strong H⁻ opacity signal in the near-infrared emission spectrum of the ultra-hot Jupiter KELT-9b
Astronomy and Astrophysics; L1. 10 citations.
46. Dymont, A. H., Yu, X., Ohno, K., Zhang, X., et al. (2022)
Cleaning Our Hazy Lens: Exploring Trends in Transmission Spectra of Warm Exoplanets
The Astrophysical Journal; 2, 90. 33 citations.
47. Kreidberg, L., Mollière, P., Crossfield, I. J. M., Thorngren, D. P., et al. (2022)
Tentative Evidence for Water Vapor in the Atmosphere of the Neptune-sized Exoplanet HD 106315c
The Astronomical Journal; 4, 124. 40 citations.
48. Chachan, Y., Dalba, P. A., Knutson, H. A., Fulton, B. J., et al. (2022)
Kepler-167e as a Probe of the Formation Histories of Cold Giants with Inner Super-Earths
The Astrophysical Journal; 1, 62. 23 citations.
49. Dalba, P. A., Kane, S. R., Dragomir, D., Villanueva, S., et al. (2022)
The TESS-Keck Survey. VIII. Confirmation of a Transiting Giant Planet on an Eccentric 261 Day Orbit with the Automated Planet Finder Telescope
The Astronomical Journal; 2, 61. 29 citations.
50. Dang, L., Bell, T. J., Cowan, N. B., Thorngren, D., et al. (2022)
Thermal Phase Curves of XO-3b: An Eccentric Hot Jupiter at the Deuterium Burning Limit
The Astronomical Journal; 1, 32. 10 citations.
51. Wahl, S. M., Thorngren, D., Lu, T., Militzer, B. (2021)
Tidal Response and Shape of Hot Jupiters
The Astrophysical Journal; 2, 105. 13 citations.
52. Dalba, P. A., Kane, S. R., Li, Z., MacDougall, M. G., et al. (2021)
Giant Outer Transiting Exoplanet Mass (GOT 'EM) Survey. II. Discovery of a Failed Hot Jupiter on a 2.7 Yr, Highly Eccentric Orbit
The Astronomical Journal; 4, 154. 16 citations.
53. Hobson, M. J., Brahm, R., Jordán, A., Espinoza, N., et al. (2021)
A Transiting Warm Giant Planet around the Young Active Star TOI-201
The Astronomical Journal; 5, 235. 44 citations.

54. Baxter, C., Désert, J.-M., Tsai, S.-M., Todorov, K. O., et al. (2021)
Evidence for disequilibrium chemistry from vertical mixing in hot Jupiter atmospheres. A comprehensive survey of transiting close-in gas giant exoplanets with warm-Spitzer/IRAC
Astronomy and Astrophysics; A127. 44 citations.
55. Piaulet, C., Benneke, B., Rubenzahl, R. A., Howard, A. W., et al. (2021)
WASP-107b's Density Is Even Lower: A Case Study for the Physics of Planetary Gas Envelope Accretion and Orbital Migration
The Astronomical Journal; 2, 70. 82 citations.
56. Mikal-Evans, T., Crossfield, I. J. M., Benneke, B., Kreidberg, L., et al. (2021)
Transmission Spectroscopy for the Warm Sub-Neptune HD 3167c: Evidence for Molecular Absorption and a Possible High-metallicity Atmosphere
The Astronomical Journal; 1, 18. 34 citations.
57. Mayorga, L. C., Charbonneau, D., Thorngren, D. P. (2020)
Reflected Light Observations of the Galilean Satellites from Cassini: A Test Bed for Cold Terrestrial Exoplanets
The Astronomical Journal; 5, 238. 7 citations.
58. Díaz, M. R., Jenkins, J. S., Feng, F., Butler, R. P., et al. (2020)
The Magellan/PFS Exoplanet Search: a 55-d period dense Neptune transiting the bright ($V = 8.6$) star HD 95338
Monthly Notices of the Royal Astronomical Society; 4, 4330. 23 citations.
59. Komacek, T. D., Thorngren, D. P., Lopez, E. D., Ginzburg, S. (2020)
Reinflation of Warm and Hot Jupiters
The Astrophysical Journal; 1, 36. 31 citations.
60. Movshovitz, N., Fortney, J. J., Mankovich, C., Thorngren, D., et al. (2020)
Saturn's Probable Interior: An Exploration of Saturn's Potential Interior Density Structures
The Astrophysical Journal; 2, 109. 37 citations.
61. Vissapragada, S., Jontof-Hutter, D., Shporer, A., Knutson, H. A., et al. (2020)
Diffuser-assisted Infrared Transit Photometry for Four Dynamically Interacting Kepler Systems
The Astronomical Journal; 3, 108. 61 citations.
62. Teske, J. K., Thorngren, D., Fortney, J. J., Hinkel, N., et al. (2019)
Do Metal-rich Stars Make Metal-rich Planets? New Insights on Giant Planet Formation from Host Star Abundances
The Astronomical Journal; 6, 239. 49 citations.
63. Wallack, N. L., Knutson, H. A., Morley, C. V., Moses, J. I., et al. (2019)
Investigating Trends in Atmospheric Compositions of Cool Gas Giant Planets Using Spitzer Secondary Eclipses
The Astronomical Journal; 6, 217. 27 citations.
64. Espinoza, N., Fortney, J. J., Miguel, Y., Thorngren, D., et al. (2017)
Metal Enrichment Leads to Low Atmospheric C/O Ratios in Transiting Giant Exoplanets
The Astrophysical Journal; 1, L9. 94 citations.
65. Morley, C. V., Knutson, H., Line, M., Fortney, J. J., et al. (2017)
Forward and Inverse Modeling of the Emission and Transmission Spectrum of GJ 436b: Investigating Metal Enrichment, Tidal Heating, and Clouds
The Astronomical Journal; 2, 86. 120 citations.
66. Szydagis, M., Fyhrie, A., Thorngren, D., Tripathi, M. (2013)
Enhancement of NEST capabilities for simulating low-energy recoils in liquid xenon
Journal of Instrumentation; 10, C10003. 83 citations.